

# k-th order statistic Index Preservation under Bounded Score Perturbations

Research Architecture Runtime  
operator/kth-order-statistic

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## Abstract

The kth order statistic operator selects a unique strict  $k$ -th order-statistic index. When every pairwise score gap exceeds  $2\varepsilon$ , that index is invariant under  $\|\delta\|_\infty \leq \varepsilon$ . Lean: `LEAN_FULL` (reduction of `OrderStat.KthMargin`).

**This theorem is:** Reduction.

## 1 Problem

The `kth-order-statistic` operator on scores  $s \in \mathbb{R}^n$  ( $n \geq 2$ ) returns the unique index  $i$  that is the strict  $k$ -th smallest coordinate of  $s$  (operator-specific choice of  $k$ ). The selected object is this index under a unique strict-order presentation.

## 2 Stability notion

Perturbations are additive  $\delta$  with  $\|\delta\|_\infty \leq \varepsilon$ . Stability: if  $i$  is the unique strict  $k$ -th smallest and `AllGapsExceed`( $s, 2\varepsilon$ ), then  $i$  remains the unique strict  $k$ -th smallest of  $s + \delta$ . Sharpness: a rival within distance  $2\varepsilon$  admits an admissible tying adversary.

## 3 Definitions

**Definition 1** (Strict  $k$ -th smallest). *Index  $i$  is strict  $k$ -th smallest if  $\#\{j : s_j < s_i\} = k$  and  $s_j = s_i \Rightarrow j = i$ .*

**Definition 2** (All-gaps).  $|s_p - s_q| > g$  for all  $p \neq q$ .

**Assumptions.**  $n \geq 2$ ,  $\varepsilon \geq 0$ , unique strict  $k$ -th index, and (for invariance) all pairwise gaps  $> 2\varepsilon$ .

## 4 Theorem

**Theorem 3** (Invariance). *If  $i$  is strict  $k$ -th smallest and `AllGapsExceed`( $s, 2\varepsilon$ ), then for every  $\|\delta\|_\infty \leq \varepsilon$ ,  $i$  remains strict  $k$ -th smallest for  $s + \delta$ .*

**Theorem 4** (Sharpness). *If some  $j \neq i$  has  $|s_i - s_j| \leq 2\varepsilon$ , an admissible midpoint  $\delta$  destroys uniqueness.*

## 5 Intuition

An  $\ell_\infty$  adversary moves any two scores by at most  $2\varepsilon$  relative to each other. Gaps larger than  $2\varepsilon$  cannot reverse or tie; gaps of size at most  $2\varepsilon$  admit a midpoint collision.

## 6 Examples

**Example 5.** Scores  $(1, 4, 7)$ ,  $k = 1$ ,  $\varepsilon = 1$ : gaps  $3, 6, 3$  all exceed  $2$ , so the middle index is stable. For  $\varepsilon = 2$ , the gap  $3 \leq 4$  admits a midpoint tie between the first two coordinates.

## 7 Proof

Let  $s' = s + \delta$  with  $\|\delta\|_\infty \leq \varepsilon$ .

**Pairwise orders.** If  $s_p < s_q$  and  $s_q - s_p > 2\varepsilon$ , then  $s'_p - s'_q \leq -(s_q - s_p) + 2\varepsilon < 0$ , so  $s'_p < s'_q$ . The converse is symmetric. Hence  $s_p < s_q \Leftrightarrow s'_p < s'_q$  whenever  $|s_p - s_q| > 2\varepsilon$ ; under all-gaps this holds for every distinct pair.

**Count.** Taking  $q = i$  yields  $\#\{j : s_j < s_i\} = \#\{j : s'_j < s'_i\}$ . If the left side equals  $k$ , so does the right.

**Uniqueness.** If  $s'_j = s'_i$  and  $j \neq i$ , all-gaps give  $|s_j - s_i| > 2\varepsilon$ , hence  $s'_j \neq s'_i$ , contradiction. Thus  $i$  remains the unique strict  $k$ -th index (Theorem 3).

**Sharpness.** For  $j \neq i$  with  $|s_i - s_j| \leq 2\varepsilon$ , set  $\delta_i = -(s_i - s_j)/2$ ,  $\delta_j = (s_i - s_j)/2$ , and  $\delta_t = 0$  otherwise. Then  $\|\delta\|_\infty \leq \varepsilon$  and  $s'_i = s'_j$ , so uniqueness fails (Theorem 4).

## 8 Formal statement

Lean module: `Research.Operators.KthOrderStatistic.Preservation`. Source file: `lean/Research/Operators`  
Theorems: `kth_order_statistic_margin_invariance`, `kth_order_statistic_margin_sharpness`.  
Definitional reduction of `Research.Operators.OrderStat.KthMargin`. Certificate: `lean/certificates/kth-ord`  
Status: LEAN\_FULL on Mathlib  $\mathbb{R}$  (REAL\_MATHLIB).

## 9 Proof dependencies

- Reduces to `Research.Operators.OrderStat.KthMargin`.
- Uses the  $\ell_\infty$  relative-gap bound  $|\delta_p - \delta_q| \leq 2\varepsilon$ .
- Midpoint collision adversary.
- No other operator theorems required.

## 10 Consequences

Operators reducing to the same  $k$ -th core include `quantile`, `median`, `percentile`, `kth-order-statistic`. The `kth-order-statistic` theorem is the stability certificate for unique strict order-statistic selection under bounded score noise.